

Optimal TCP and Robot Base Placement for a Set of Complex Continuous Paths

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Abstract—The robot base placement of an industrial robot in flexible production lines is crucial due to the limited workspace of robots, in particular for complex continuous paths that change frequently. Costly and time-consuming repositioning of the robot can be avoided by merely adapting the tool center point (TCP) of the robot, which is the focus of this work. To this end, an algorithm for the optimal TCP placement for a set of tool paths is proposed. This algorithm is based on a fast joint-space path planner which is capable of moving through kinematic singularities and takes into account wide turning ranges of individual robot axes. Furthermore, the proposed concept also applies to the optimal robot base placement. The feasibility of the approach is demonstrated for a trim application in shoe production for a set of 44 complex continuous tool paths.

I. INTRODUCTION

While designing a production cell with an industrial robot, the position and orientation, i.e. the pose, of the robot base relative to the workpiece has to be chosen appropriately. In a flexible robot work cell, however, products and robot tool paths may change frequently, possibly exceeding the capabilities of the work cell. Instead of adapting the complete setup or remounting the robot, which is a costly and time-consuming process, it is often sufficient to merely adapt the tool center point (TCP) of the robot using simple mechanical adapters. In this work, we focus on optimal placement of the TCP, which gained little attention in the literature so far.

Most works in the literature deal with base pose optimization only, considering different properties of the robot, i.e. kinematics [1], dynamics [2], manipulability [3], stiffness [4], and time-optimality [5]. For example, the stiffness and deformation of individual axes are considered for the base placement in milling tasks [6] to satisfy the high accuracy requirements. In some applications, i.e. mostly for milling and welding tasks, continuous end-effector paths are investigated. In [7] and [8], the elastostatic model of the robot is used to find the optimal workpiece placement for continuous paths by minimizing the distance between the robot tool and its desired path. In [9], genetic algorithms are used that consider kinematic and dynamic manipulability and milling forces for a given trajectory. A fast search for a feasible workpiece pose is shown in [10] for a 7 degrees of freedom (DOF) robot with a tool axis redundancy, by successively considering additional constraints in an optimization problem. Other

works concerned with robot base placement only consider individual poses instead of continuous paths, e.g., for spot welding [11], [12], short seam welding [13] or pick-and-place tasks using mobile robots [14]. Further works are focused on capability and reachability maps which are mostly used to position robots in grasping tasks, e.g., [15], [16] and [17].

For most applications, industrial robots with 6-DOF, a spherical wrist and axis ranges wider than 360° are used. Therefore, more than the usual 8 inverse kinematics solutions, which result in the same end-effector pose, have to be considered. In [18], the feasibility of the whole path is investigated while focusing mainly on cycle times. The work [19] combines two optimization loops, where in the inner loop, a redundant DOF is used to avoid singularities and the dynamical limits and in the outer loop, a particle swarm optimization searches for the global optimal robot base pose. Although wide axis ranges are considered in both works, the same solution of the inverse kinematics is used throughout a single path execution.

A survey of sampling-based path planning methods which systematically incorporate constraints is given in [20]. The graph-based Descartes algorithm described in [21] considers movements through multiple inverse kinematic solutions. However, this planner only outputs a single solution for each path planning problem and exhibits comparably long computation times. More advanced path planning algorithms like [22] and [23] cover a wider range of applications, but also suffer from higher computational costs. Hence, they could not be used for our purpose.

In this work, a fast joint-space path planner is proposed which considers all feasible kinematic solutions arising from wide turning ranges of robot axes. Additionally, this planner is capable to generate multiple joint-space paths besides moving through kinematic singularities. Based on the output of the joint-space path planner, the optimal TCP is determined as the solution of an optimization problem, in which the joint-space motions and robot limits are considered. To the best of the authors' knowledge, such an approach has not been proposed in the literature yet. This work shows that adapting the TCP for a misplaced robot can avoid repositioning of the robot base. Furthermore, not only one path is considered, but a set of paths is taken as input for the optimization. The robot cell is only productive if the robot can execute all desired paths. The optimization problem is formulated in a general way, such that it is also applicable to optimal robot base placement.

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This paper is organized as follows. In Section II, the kinematic model of a general robot is introduced and the inverse kinematics is summarized. The underlying joint-space path planner is proposed in Section III. The optimization algorithm is stated in Section IV, where the optimization criteria are described in detail. In Section V, the proposed algorithm is applied to a trim application, where the optimal robot base and the TCP placement are calculated for a set of 44 complex continuous paths. Finally, conclusions are drawn in Section VI.

II. MATHEMATICAL ROBOT MODEL

In this section, the mathematical model of a general robot with f DOF is introduced [24]. For the actual application in Section V, the 6-DOF robot KUKA CYBERTECH KR8 R1620 with a spherical wrist is used, see Fig. 2.

A. Forward Kinematics

The forward kinematics of a robot with f DOF describes the end-effector frame $\mathcal{E}$ w.r.t. the robot base frame $\mathcal{B}$ in the form

$$\mathbf{y}_{\mathcal{B}}^{\mathcal{E}} = \begin{bmatrix} \mathbf{p}_{\mathcal{B}}^{\mathcal{E}} \\ \phi_{\mathcal{B}}^{\mathcal{E}} \end{bmatrix} = \mathbf{h}(\mathbf{q}) \quad , \quad (1)$$

using the Cartesian position $(\mathbf{p}_{\mathcal{B}}^{\mathcal{E}})^T = [x_{\mathcal{B}}^{\mathcal{E}} \ y_{\mathcal{B}}^{\mathcal{E}} \ z_{\mathcal{B}}^{\mathcal{E}}] \in \mathbb{R}^3$ and the EULER angles $(\phi_{\mathcal{B}}^{\mathcal{E}})^T = [\phi_{\mathcal{B},z}^{\mathcal{E}} \ \phi_{\mathcal{B},y}^{\mathcal{E}} \ \phi_{\mathcal{B},x}^{\mathcal{E}}] \in \mathbb{R}^3$ in TAIT-BRYAN convention, i.e. a sequence of rotations around the axes z , y and x . Note that also a lower-dimensional output space may be chosen in (1). The vector $\mathbf{q}^T = [q_1 \ \dots \ q_f]$ consists of the robot joint angles, which are also called a joint configuration. Equivalently, the end-effector pose $\mathbf{y}_{\mathcal{B}}^{\mathcal{E}}$ is written as a homogeneous transformation in the form

$$\mathbf{H}_{\mathcal{B}}^{\mathcal{E}}(\mathbf{q}) = \begin{bmatrix} \mathbf{R}_{\mathcal{B}}^{\mathcal{E}}(\mathbf{q}) & \mathbf{p}_{\mathcal{B}}^{\mathcal{E}}(\mathbf{q}) \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{\mathcal{B}}^{\mathcal{E}}(\phi_{\mathcal{B}}^{\mathcal{E}}) & \mathbf{p}_{\mathcal{B}}^{\mathcal{E}} \\ \mathbf{0} & 1 \end{bmatrix} \quad , \quad (2)$$

with the Cartesian position $\mathbf{p}_{\mathcal{B}}^{\mathcal{E}}$ and the rotation matrix $\mathbf{R}_{\mathcal{B}}^{\mathcal{E}} \in \text{SO}(3)$, which is computed by using $\phi_{\mathcal{B}}^{\mathcal{E}}$ from (1). In the remainder of this work, the notations $\mathbf{y}_{\mathcal{A}}^{\mathcal{B}}$ and $\mathbf{H}_{\mathcal{A}}^{\mathcal{B}}$ according to (1) and (2) are employed for all transformations between any frames $\mathcal{A}$ and $\mathcal{B}$.

B. Inverse Kinematics

The inverse function of $\mathbf{h}(\mathbf{q})$ in (1) is the inverse kinematics of a robot, formally written as

$$Q = \mathbf{h}^{-1}(\mathbf{y}_{\mathcal{B}}^{\mathcal{E}}) \quad , \quad (3)$$

where Q is the set of all possible solutions $\mathbf{q}_k$ and k is the solution index. The solution of (3) requires the use of trigonometric functions like $\text{atan2}(y, x): \mathbb{R} \times \mathbb{R} \rightarrow [-\pi, \pi]$, which considers the four quadrants of the input arguments y and x [24]. Robot joints with a wide turning range beyond $[-\pi, \pi]$ provide additional feasible solutions, which are found by adding $\pm 2\pi$ accordingly.

III. JOINT-SPACE PATH PLANNER

The algorithm for the optimal TCP placement in Section IV is based on the results of a joint-space path planner which is introduced in this section. In the following, the TCP frame is denoted by $\mathcal{T}$. The objective of the path planner is to find all feasible continuous joint-space paths P for a given task-space path H , also denoted as tool path. The task-space path H is defined as a sequence containing the homogeneous transformations $\mathbf{H}_{\mathcal{B},d}^{\mathcal{T},i}, i = 1, \dots, n$. The joint-space path planner is formulated in Algorithm 1 in the following.

Algorithm 1 Joint-space path planner

Require: $\mathbf{H}_{\mathcal{B},d}^{\mathcal{T},i}, i = 1, \dots, n$
INITIALIZE:
for $i = 1$ **to** n **do**
 $\mathbf{H}_{\mathcal{B}}^{\mathcal{E},i} \leftarrow \mathbf{H}_{\mathcal{B},d}^{\mathcal{T},i} (\mathbf{H}_{\mathcal{E}}^{\mathcal{T}})^{-1}$
 $Q^i \leftarrow \mathbf{h}^{-1}(\mathbf{y}_{\mathcal{B}}^{\mathcal{E},i})$
 $L^i \leftarrow \text{feasibleSolutionIndices}(Q^i)$
end for
SEARCH CONTINUOUS JOINT-SPACE PATHS:
 $C \leftarrow L^1$
 $\forall c \in C : P_c \leftarrow \{\mathbf{q}_c^1\}$
for $i = 2$ **to** n **do**
for all $P_c, c \in C$ **do**
 $l_{\min} \leftarrow \arg \min_{l \in L^i} \|\mathbf{q}_l^i - \text{lastElement}(P_c)\|_2$
if $\|\mathbf{q}_{l_{\min}}^i - \text{lastElement}(P_c)\|_2 < T$ **then**
 $P_c \leftarrow P_c \cup \{\mathbf{q}_{l_{\min}}^i\}$
else
 $C \leftarrow C \setminus c$
end if
end for
end for

First, in the initialization phase of Algorithm 1, the inverse kinematics from Section II-B is applied to each pose $\mathbf{H}_{\mathcal{B},d}^{\mathcal{T},i}$ of the path using

$$\mathbf{H}_{\mathcal{B}}^{\mathcal{E},i} = \mathbf{H}_{\mathcal{B},d}^{\mathcal{T},i} (\mathbf{H}_{\mathcal{E}}^{\mathcal{T}})^{-1} \quad (4a)$$

$$Q^i = \mathbf{h}^{-1}(\mathbf{y}_{\mathcal{B}}^{\mathcal{E},i}) \quad , \quad (4b)$$

for $i = 1, \dots, n$ to obtain the sets Q^i containing $s \in \mathbb{N}$ inverse kinematic solutions $\mathbf{q}_k^i$ for all poses $\mathbf{y}_{\mathcal{B}}^{\mathcal{E},i}$. Let L^i denote the index set of feasible solutions of Q^i , i.e. all inverse kinematic solutions which satisfy the kinematic constraints like mechanical joint limits. Note that if the DOF of the robot are larger than the dimension of the output space in (1) the sets Q^i have to be sampled accordingly [21]. Then, starting with $c \in L^1$, all possible sequences of feasible continuous joint-space paths P_c are determined. The first element of the sequence $P_c, c \in L^1$, corresponds to the inverse kinematics solution $\mathbf{q}_c^1$. The second element of the sequence $P_c, c \in L^1$, is obtained by minimizing the Euclidean distance $\|\mathbf{q}_l^2 - \mathbf{q}_c^1\|_2$ for all $l \in L^2$, where $l_{\min}$ refers to the index of the solution with minimum distance. If $\|\mathbf{q}_{l_{\min}}^2 - \mathbf{q}_c^1\|_2 < T$, with a user-defined threshold T , then the second element

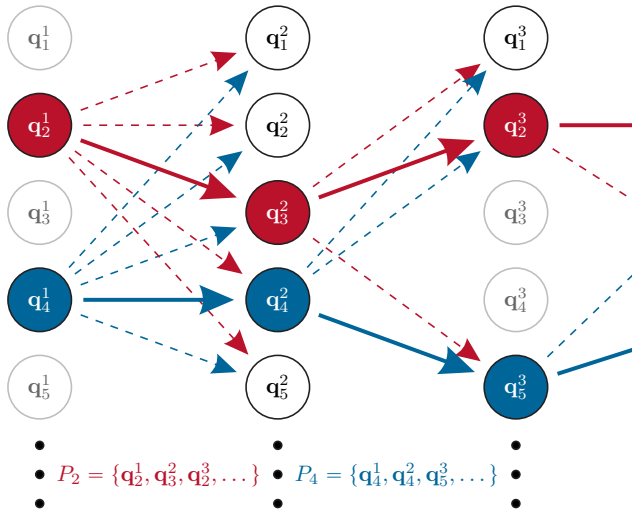


Fig. 1. Example of the path planner searching for continuous paths, where grey nodes represent non-feasible joint configurations. Thereby, solid arrows represent transitions with the shortest Euclidean distance, while dashed arrows represent discarded transitions between feasible joint configurations.

of the sequence P_c is $\mathbf{q}_{l_{\min}}^2$, otherwise P_c is discarded and a new index set C of feasible continuous joint-space paths is defined as $C = L^1 \setminus c$. This procedure is now sequentially applied to all further path points $i = 3, \dots, n$. With this approach, multiple continuous joint-space paths P_c , $c \in C$, are found for a given task-space path H . This joint-space path planner is capable of considering movements through kinematic singularities and wide turning axis ranges, since every possible sequence of inverse kinematic solutions is examined.

A visual representation of the search for continuous joint-space paths is shown in Fig. 1. In this figure, light grey nodes represent non-feasible solutions due to kinematic constraints, while black nodes represent feasible joint configurations. The shortest Euclidean distances are indicated as arrows with solid lines and the dashed lines represent discarded transitions.

IV. OPTIMAL TCP PLACEMENT

In this section, the algorithm to determine the optimal pose $\mathbf{y}_{\mathcal{E}}^T$ of the TCP frame $\mathcal{T}$ w.r.t. the end-effector frame $\mathcal{E}$ is described. By simply adapting the TCP of the robot using an optimization-based approach, redesigning the robot work cell and repositioning of the robot can be avoided. First, in Section IV-A, a set of paths is introduced as input for the optimization. Second, the objective function and the optimization criteria are presented in Section IV-B and third, the optimization algorithm is specified in detail in Section IV-C. Finally, in Section IV-D, the algorithm is adapted for optimal robot base placement.

A. Set of Paths

The set of tool paths $\{H_1, \dots, H_m\}$ is considered as input for the algorithm. The joint-space path planning algorithm from Section III is executed for all tool paths H_j , $j = 1, \dots, m$, for a given TCP pose $\mathbf{y}_{\mathcal{E}}^T$, resulting in the feasible joint-space paths $P^j = \{P_c^j \mid c \in C_j\}$, $j = 1, \dots, m$, for

all tool paths. Note that C_j is the index set for the feasible solutions of the corresponding path j . Each tool path H_j consists of n_j poses.

B. Objective Function and Optimization Criteria

The objective function $f(\mathbf{y}_{\mathcal{E}}^T)$ computes a scalar measure for a given TCP pose $\mathbf{y}_{\mathcal{E}}^T$. The results of the joint-space path planner from Section III are used as qualitative and quantitative measure in the objective function of the optimization. Note that the optimization problem is formulated as a maximization problem.

The objective function $f(\mathbf{y}_{\mathcal{E}}^T)$ consists of four terms

$$f(\mathbf{y}_{\mathcal{E}}^T) = \frac{a_1 f_1(\mathbf{y}_{\mathcal{E}}^T) + a_2 f_2(\mathbf{y}_{\mathcal{E}}^T) + a_3 f_3(\mathbf{y}_{\mathcal{E}}^T) + a_4 f_4(\mathbf{y}_{\mathcal{E}}^T)}{a_1 + a_2 + a_3 + a_4}, \quad (5)$$

where the coefficients a_t are manually tuned weights of the individual terms $f_t(\mathbf{y}_{\mathcal{E}}^T)$, $t = 1, \dots, 4$. The individual terms $f_t(\mathbf{y}_{\mathcal{E}}^T)$ are constructed as follows:

- The term $f_1(\mathbf{y}_{\mathcal{E}}^T)$ accounts for the number of feasible inverse kinematic solutions $s_{i,j}$ for every path point $i = 1, \dots, n_j$ of every tool path $j = 1, \dots, m$, i.e.

$$f_1(\mathbf{y}_{\mathcal{E}}^T) = \frac{1}{sm} \sum_{j=1}^m \frac{1}{n_j} \sum_{i=1}^{n_j} s_{i,j} \quad (6)$$

Note that the pre-factor in (6) normalizes the function. The constant s denotes the maximum number of inverse kinematics solutions for the robot. The term (6) locally generates a gradient in a meaningful direction and, thus, local minima are avoided in the objective function (5).

- The function $f_2(\mathbf{y}_{\mathcal{E}}^T)$ corresponds to the number of feasible continuous solutions for each tool path $j = 1, \dots, m$

$$f_2(\mathbf{y}_{\mathcal{E}}^T) = \frac{1}{sm} \sum_{j=1}^m |C_j| \quad (7)$$

where $|C_j|$ denotes the cardinality of the set C_j . The higher the objective term (7), the more tool paths are continuously executable by the robot.

- By including $f_3(\mathbf{y}_{\mathcal{E}}^T)$ in the objective function, the joint movement between consecutive path points is minimized for continuous paths. The objective term $f_3(\mathbf{y}_{\mathcal{E}}^T)$ is defined as

$$f_3(\mathbf{y}_{\mathcal{E}}^T) = \frac{1}{m} \sum_{j=1}^m \frac{1}{(n_j-1)|C_j|} \sum_{c \in C_j} \sum_{i=1}^{n_j-1} \|\mathbf{q}^{i+1} - \mathbf{q}^i\|_2^{-1}, \quad (8)$$

where $\mathbf{q}^i$ is the i^{th} element of the sequence P_c^j . Again, the pre-factors normalize the objective term (8).

- A robot setup is more flexible, if it is also able to execute paths, for which the TCP was not optimized. Hence, the last term $f_4(\mathbf{y}_{\mathcal{E}}^T)$ considers the reserves of the joint angles to the respective mechanical joint limits $q_{h,\min}$ and $q_{h,\max}$, $h = 1, \dots, f$ during the robot motion and is given by

$$f_4(\mathbf{y}_{\mathcal{E}}^T) = \frac{1}{m} \sum_{j=1}^m \frac{1}{n_j|C_j|} \sum_{c \in C_j} \sum_{i=1}^{n_j} \|\bar{\mathbf{q}}(\mathbf{q}^i)\|_2^{-1} \quad (9)$$

with $\mathbf{q}^i \in P_c^j$, where the vector-valued function $\bar{\mathbf{q}}^T(\mathbf{q}) = [\bar{q}_1 \ \cdots \ \bar{q}_f]$ contains the entries

$$\bar{q}_h = \frac{2q_h - (q_{h,\max} + q_{h,\min})}{q_{h,\max} - q_{h,\min}}, \quad h = 1, \dots, f. \quad (10)$$

C. Optimization Algorithm

Using the objective function (5), the optimization problem for the optimal TCP placement is formulated as

$$\mathbf{y}_{\mathcal{E},\text{opt}}^T = \arg \max_{\mathbf{y}_{\mathcal{E}}^T \in \mathbb{R}^6} f(\mathbf{y}_{\mathcal{E}}^T) \quad (11a)$$

$$\text{s.t. } \mathbf{y}_{\mathcal{E},\min}^T \leq \mathbf{y}_{\mathcal{E}}^T \leq \mathbf{y}_{\mathcal{E},\max}^T. \quad (11b)$$

Note that $f(\mathbf{y}_{\mathcal{E}}^T)$ is not continuously differentiable. Furthermore, the path planning is computationally quite expensive, in particular if a large set of paths is considered. Therefore, a surrogate optimization approach is chosen. Specifically, the MATLAB implementation SURROGATEOPT is used to solve (11), which uses an approximation of the objective function to search for the global optimum [25]. Additionally, the parallel nature of this algorithm is utilized to reduce the computation time.

D. Robot Base Placement

By considering a new optimization variable, the problem (11) is adapted in a straightforward way for optimal robot base placement, which is described in this section. To this end, the offset frame $\mathcal{O}$ is introduced and the homogeneous transformation $\mathbf{H}_{\mathcal{B},d}^{\mathcal{T},i}$ from (4) is reformulated as

$$\mathbf{H}_{\mathcal{B},d}^{\mathcal{T},i}(\mathbf{x}_{\mathcal{B}}^{\mathcal{O}}) = \mathbf{H}_{\mathcal{B}}^{\mathcal{O}}(\mathbf{x}_{\mathcal{B}}^{\mathcal{O}}) \mathbf{H}_{\mathcal{O},d}^{\mathcal{T},i}, \quad (12)$$

with the new optimization variable $(\mathbf{x}_{\mathcal{B}}^{\mathcal{O}})^T = [x_{\mathcal{B}}^{\mathcal{O}} \ y_{\mathcal{B}}^{\mathcal{O}}]$ and a constant height $z_{\mathcal{B}}^{\mathcal{O}} = 0.193\text{m}$. Note that the tool paths $\mathbf{H}_{\mathcal{O},d}^{\mathcal{T},i}$ in (12) are now specified w.r.t. the offset frame $\mathcal{O}$. The optimization problem (11) is adapted in the form

$$\mathbf{x}_{\mathcal{B},\text{opt}}^{\mathcal{O}} = \arg \max_{\mathbf{x}_{\mathcal{B}}^{\mathcal{O}} \in \mathbb{R}^2} f(\mathbf{x}_{\mathcal{B}}^{\mathcal{O}}) \quad (13a)$$

$$\text{s.t. } \mathbf{x}_{\mathcal{B},\min}^{\mathcal{O}} \leq \mathbf{x}_{\mathcal{B}}^{\mathcal{O}} \leq \mathbf{x}_{\mathcal{B},\max}^{\mathcal{O}}, \quad (13b)$$

which is again solved with the surrogate optimization approach.

V. RESULTS

In this section, the optimal placement of the TCP and the robot base is applied to a robot work cell performing a trim application. Thereby, a robot trims excess material from a shoe sole arising from injection molding, which is a common task in semi-automated shoe manufacturing. In this application, the 6-DOF robot KUKA CYBERTECH KR8 R1620 with a spherical wrist is used, see Fig. 2. The DENAVIT-HARTENBERG parameters [24] as well as the mechanical joint limits $\mathbf{q}_{\min}$ and $\mathbf{q}_{\max}$ of the robot are listed in Tab. I. Note that the range of joint 3 is constrained additionally by the workshop floor and the energy chain of the tool. The robot base and the TCP have to be placed such that all 22 trim paths for the left and the right shoe, i.e.

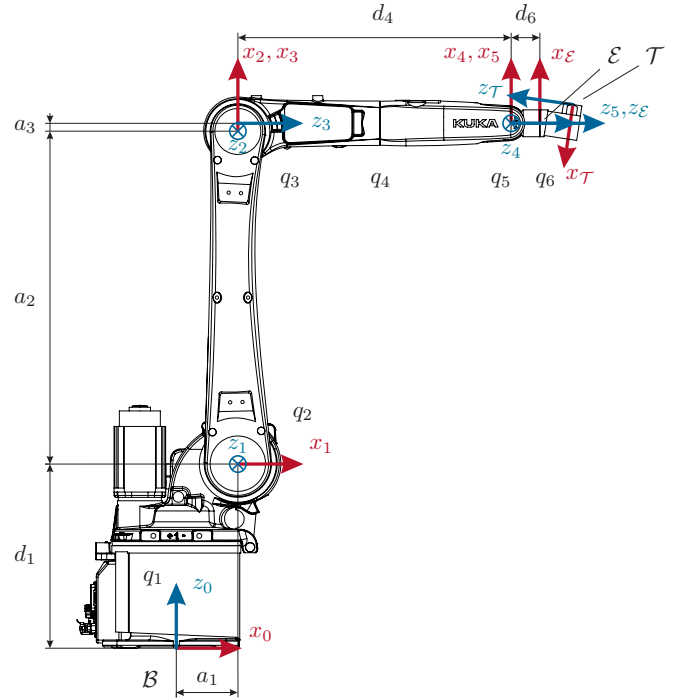


Fig. 2. Coordinate frames of the KUKA CYBERTECH KR8 R1620 with revolute joints q_h turning around the z_{h-1} axes, $h = 1, \dots, 6$.

44 paths in total, are executable. Two of the trim paths are shown in Fig. 3, where the robot base frame $\mathcal{B}$ is located at the origin. All trim paths are described in the offset frame $\mathcal{O}$. All feasible joint-space solutions for a single trim path with $n = 638$ are found in less than 100 ms using the fast joint-space path planner from Section III on a CORE i7-8700K with 16 GB RAM.

A. Optimal Robot Base Placement

In this section, the placement of the robot base for the trim application is investigated, which is analogous to workpiece placement. The homogeneous transformation $\mathbf{H}_{\mathcal{E}}^T$ in (4) is taken from the existing tool as $(\mathbf{y}_{\mathcal{E},\text{ex}}^T)^T = [-0.066\text{m} \ 0 \ 0.269\text{m} \ \pi \ 0.719\text{rad} \ -\pi]$.

To illustrate and discuss the objective function $f(\mathbf{x}_{\mathcal{B}}^{\mathcal{O}})$ for the surrogate optimization algorithm, a MONTE-CARLO simulation in the admissible range of $\mathbf{x}_{\mathcal{B}}^{\mathcal{O}}$ is performed and shown in Fig. 4. In this figure, dark blue areas represent high values of the objective function and red areas indicate low values. Furthermore, Fig. 4 reveals three distinct plateaus, originating from the objective term $f_2(\mathbf{y}_{\mathcal{E}}^T)$. For a robot base placement $\mathbf{x}_{\mathcal{B}}^{\mathcal{O}}$ in the blue area, the joint-space paths for

TABLE I
DENAVIT-HARTENBERG PARAMETERS OF THE
KUKA CYBERTECH KR8 R1620, SEE [26]

h	Θ_h	$d_h(\text{mm})$	$\vartheta_h(^{\circ})$	$a_h(\text{mm})$	$q_{h,\min/\max}(^{\circ})$
1	q_1	450	-90	150	± 170
2	q_2	0	0	810	$-185/65$
3	$q_3 - \frac{\pi}{2}$	0	-90	20	$60/163$
4	q_4	660	90	0	± 185
5	q_5	0	-90	0	± 120
6	q_6	80	0	0	± 350

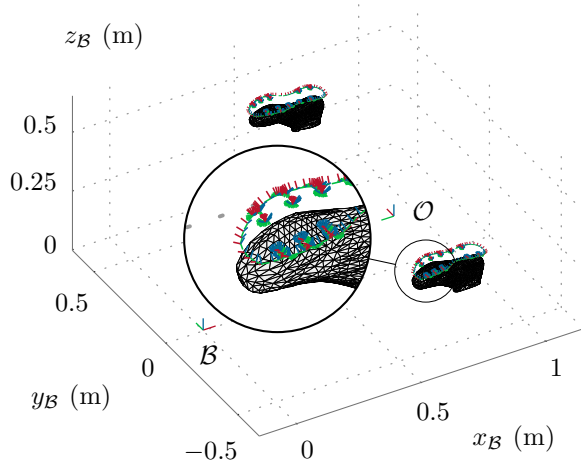


Fig. 3. Example trim paths for the left and the right shoe given in the offset frame $\mathcal{O}$.

all trim paths can be solved by the algorithm. The green area marks positions, where only a subset of trim paths can be solved and in the red area, none of the trim paths are executable by the robot.

With the boundaries $(\mathbf{x}_{B,\min}^{\mathcal{O}})^T = [0 \quad -1 \text{ m}]$ and $(\mathbf{x}_{B,\max}^{\mathcal{O}})^T = [1.5 \text{ m} \quad 1 \text{ m}]$ given by the possible workspace at the production line, the optimal robot base placement

$$(\mathbf{x}_{B,\text{opt}}^{\mathcal{O}})^T = [0.902 \text{ m} \quad 0.003 \text{ m}] \quad (14)$$

is obtained. This optimal point $\mathbf{x}_{B,\text{opt}}^{\mathcal{O}}$ is found by applying the surrogate optimization, in which the objective function (5) is evaluated about 25 times. If the robot base is placed at this optimal point $\mathbf{x}_{B,\text{opt}}^{\mathcal{O}}$, all 44 trim paths can be executed with the TCP on the robot.

For a single trim path, Fig. 5 shows a comparison of the

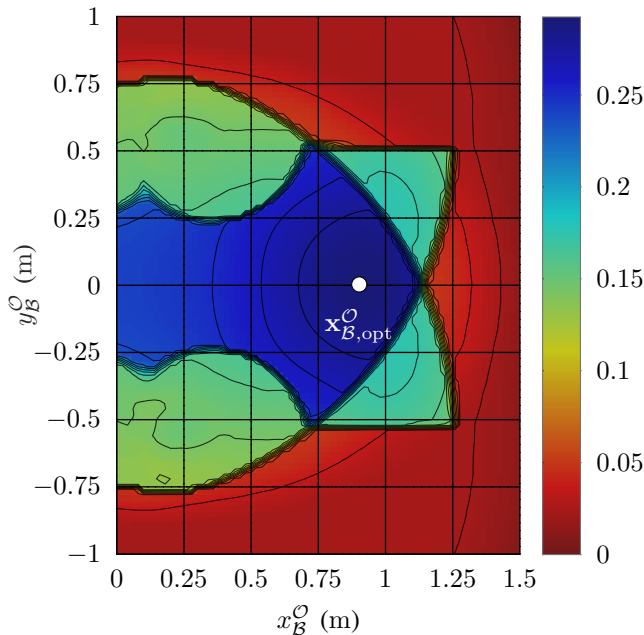


Fig. 4. Grid-based MONTE-CARLO simulation with contour lines with the optimal robot base placement $\mathbf{x}_{B,\text{opt}}^{\mathcal{O}}$.

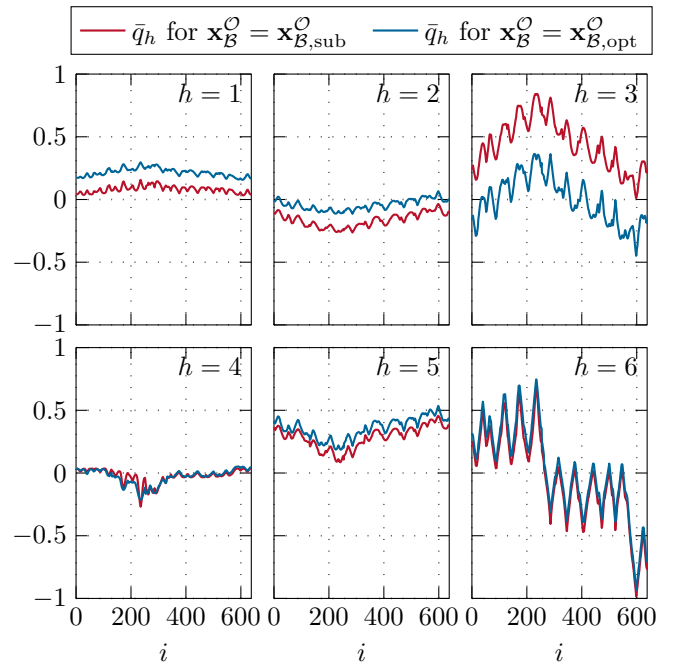


Fig. 5. Joint-space paths normalized to the respective joint limit for the optimal robot base placement $(\mathbf{x}_{B,\text{opt}}^{\mathcal{O}})^T = [0.902 \text{ m} \quad 0.003 \text{ m}]$ in blue and for the suboptimal placement $(\mathbf{x}_{B,\text{sub}}^{\mathcal{O}})^T = [0.8 \text{ m} \quad -0.4 \text{ m}]$ in red.

joint-space paths for two different robot base placements $\mathbf{x}_B^{\mathcal{O}}$, i.e. the optimal point $\mathbf{x}_{B,\text{opt}}^{\mathcal{O}}$ from (14) shown in blue and a suboptimal point $(\mathbf{x}_{B,\text{sub}}^{\mathcal{O}})^T = [0.8 \text{ m} \quad -0.4 \text{ m}]$ depicted in red. Thereby, the joint angles are normalized to $[-1, 1]$ in the range $[\mathbf{q}_{\min}, \mathbf{q}_{\max}]$ according to Tab. I. Although all desired trim paths can be executed for both placements, a distinct difference is clearly visible in the joint-space paths. In particular for joint q_3 , the joint motions for the suboptimal placement provide less reserves to the joint limits of the robot, cf. Tab. I. This is taken into account by the objective function $f_4(\mathbf{y}_{\mathcal{E}}^T)$ from Section IV-B, which penalizes joint-space path points close to the joint limits. Furthermore, Fig. 5 reveals that the wide turning range of joint 6 is necessary to perform the continuous motion along the complex trim path.

B. Optimal TCP Placement

In this section, the optimal TCP placement is investigated for all 44 trim paths considering the suboptimal robot placement $(\mathbf{x}_B^{\mathcal{O}})^T = [0.6 \text{ m} \quad -0.4 \text{ m}]$. Note that with this placement, only a subset of all trim paths can be executed with the available tool $\mathbf{y}_{\mathcal{E},\text{ex}}^T$ introduced in Section V-A, see Fig. 4. The optimization problem (11) is solved with the boundaries $(\mathbf{y}_{\mathcal{E},\min}^T)^T = [-0.2 \text{ m} \quad -0.2 \text{ m} \quad 0.2 \text{ m} \quad -\pi \quad -\pi/2 \quad -\pi]$ and $(\mathbf{y}_{\mathcal{E},\max}^T)^T = [0.2 \text{ m} \quad 0.2 \text{ m} \quad 0.4 \text{ m} \quad \pi \quad \pi/2 \quad \pi]$, which arise from the mechanical limits for the construction of the trim tool.

The optimal TCP pose $\mathbf{y}_{\mathcal{E},\text{opt}}^T$ for the optimization problem (11) is found as

$$(\mathbf{y}_{\mathcal{E},\text{opt}}^T)^T = [-0.079 \text{ m} \quad 0.039 \text{ m} \quad 0.309 \text{ m} \quad 2.849 \text{ rad} \quad 0.805 \text{ rad} \quad 3.040 \text{ rad}] \quad (15)$$

$$= [(\mathbf{p}_{\mathcal{E},\text{opt}}^T)^T \quad (\phi_{\mathcal{E},\text{opt}}^T)^T]^T,$$

for which all trim paths can be solved and therefore a repositioning of the robot can be avoided. Figure 6 illustrates the updated MONTE-CARLO simulation from Fig. 4 with the optimal TCP pose $\mathbf{y}_{\mathcal{E},\text{opt}}^T$ from (15). In this figure, the investigated point $\mathbf{x}_B^O$ is now located in the blue area, which again refers to the area where all desired trim paths can be executed by the robot. Note that this point was located in the green area in Fig. 4 using the previous TCP $\mathbf{y}_{\mathcal{E},\text{ex}}^T$.

In order to validate this result qualitatively, the admissible area of the TCP pose is investigated by visualizing the objective function $f(\mathbf{y}_{\mathcal{E}}^T)$ from (5) for the 6-dimensional space $\mathbf{y}_{\mathcal{E}}^T$ using two 3-dimensional MONTE-CARLO simulations for the position and the orientation part separately. In Fig. 7, the MONTE-CARLO simulation of the Cartesian position $\mathbf{p}_{\mathcal{E}}^T$ is shown whereas the orientation of the TCP $\phi_{\mathcal{E}}^T$ is fixed at the found optimum $\phi_{\mathcal{E},\text{opt}}^T$. Note that only TCP positions $\mathbf{p}_{\mathcal{E}}^T$ for which all trim paths can be executed are plotted. The blue dots mark high values of the objective function and low values are drawn in red. Clearly a gradient in the objective function is visible, which guides the optimization algorithm towards the optimal position $\mathbf{p}_{\mathcal{E},\text{opt}}^T$. Similarly, a MONTE-CARLO simulation of the EULER angles $\phi_{\mathcal{E}}^T$ using the fixed optimal TCP position $\mathbf{p}_{\mathcal{E},\text{opt}}^T$ from (15) is depicted in Fig. 8. Again, only data points $\phi_{\mathcal{E}}^T$ are shown, with which the robot can execute all 44 desired trim paths. The feasible areas in Fig. 8 are more distinct compared to Fig. 7 for the TCP position $\mathbf{p}_{\mathcal{E}}^T$. Note that the apparent four separated areas are in fact connected due to the periodicity of the EULER angles $\phi_{\mathcal{E},z}^T$ and $\phi_{\mathcal{E},x}^T$. Therefore, the objective function $f(\mathbf{y}_{\mathcal{E}}^T)$ exhibits a distinct maximum which is found by the surrogate optimization.

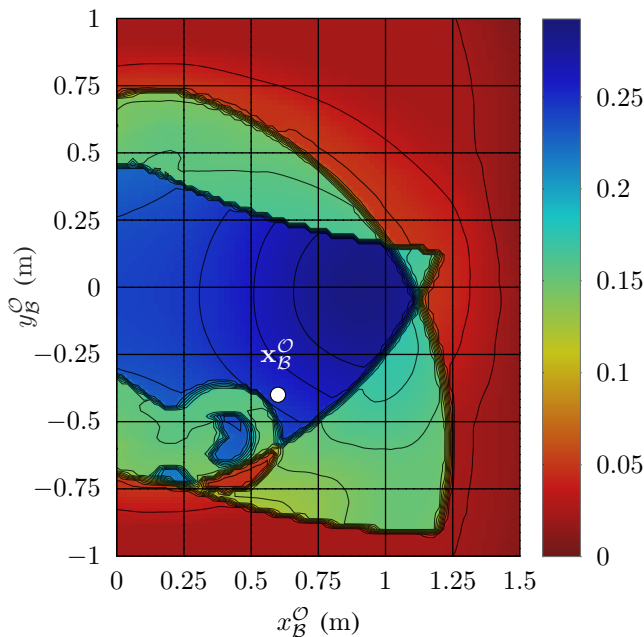


Fig. 6. Grid-based MONTE-CARLO simulation with contour lines using the TCP pose $\mathbf{y}_{\mathcal{E},\text{opt}}^T$.

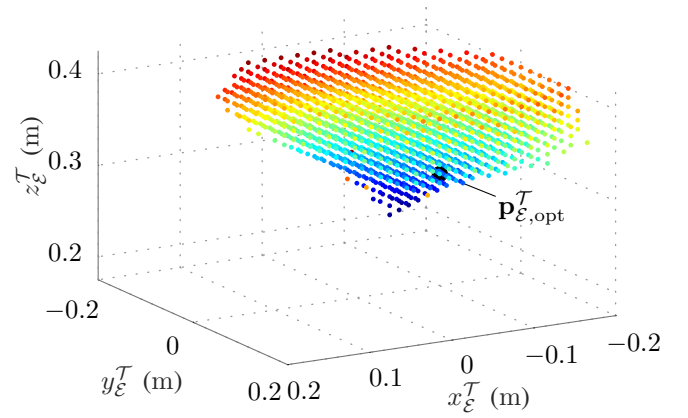


Fig. 7. Grid-based MONTE-CARLO simulation of the TCP position $\mathbf{p}_{\mathcal{E}}^T$ with fixed orientation $\phi_{\mathcal{E},\text{opt}}^T$ and high objective term values in blue.

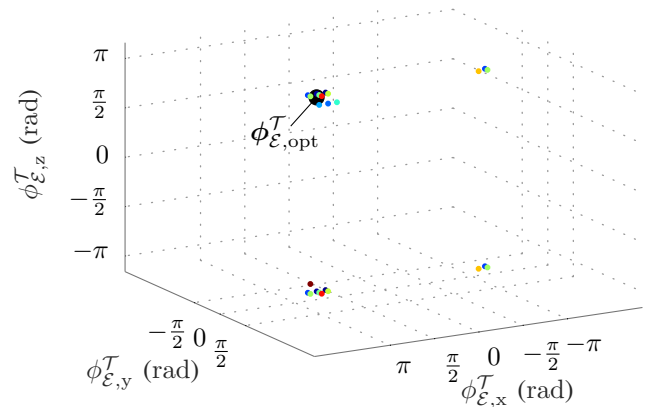


Fig. 8. Grid-based MONTE-CARLO simulation of the TCP orientation $\phi_{\mathcal{E}}^T$ with fixed position $\mathbf{p}_{\mathcal{E},\text{opt}}^T$ and high objective term values in blue.

VI. CONCLUSIONS

In order to improve the flexibility of robot work cells, an algorithm for the optimal TCP and robot base placement for a given set of complex task-space paths is proposed. To this end, a fast joint-space path planner is developed which calculates all feasible continuous joint-space paths. This planner takes into account all possible inverse kinematics solutions, including axes with wide turning ranges and moving through singular points. The optimization-based placement of the TCP and the robot base is performed using a surrogate optimization and systematically considers the distance to the respective joint limits. The proposed algorithm is applied in a trim application in shoe manufacturing to find the optimal robot base placement and the optimal TCP placement in case of a suboptimally placed robot base. This work shows that in some cases it is sufficient to adapt the TCP of the robot tool instead of redesigning the robot work cell. Additionally, placing the robot base optimally increases the reserves to the respective robot joint limits. Future work focuses on incorporating collision avoidance and the dynamic properties of the robot into the joint-space path planner for applications in constrained environments.

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